Towards digital approach for managing request for information (RFI) in construction projects: a literature review

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Abstract

Purpose – Request for information (RFI) documents play a pivotal role in seeking clarifications in construction projects. However, perceived as inevitable "non-value adding" tasks, they harbour risks like schedule delays and increased project costs, underlining the importance of strategic RFI management in construction projects. Despite this, a lack of literature dissecting RFI processes impedes a full understanding of their intricacies and impacts. This study aims to bridge the gap through a comprehensive literature review, delving into RFI intricacies and implications, while emphasising the necessity for strategic RFI management to prevent project risks.

Design/methodology/approach – This research study systematically reviews RFI-related papers published between 2000 and 2023. Accordingly, the review discusses key themes related to RFI management, yielding best practices for industry stakeholders and highlighting research directions and gaps in the body of knowledge.

Findings – Present RFI management platforms exhibit deficiencies and lack analytics essential for streamlined RFI processing. Complications arise in building information modelling (BIM)-enabled projects due to software disparities and interoperability hurdles. The existing body of knowledge heavily relies on manual content analysis, an impractical approach for the construction industry. The proposed research direction involves automated comprehension of unstructured RFI content using advanced text mining and natural language processing techniques, with the potential to greatly elevate the efficiency of RFI processing.

Originality/value – The study extends the RFI literature by providing novel insights into the problemetisation with the RFI process, offering a holistic understanding and best practices to minimise adverse effects. Additionally, the paper synthesises RFI processes in traditional and BIM-enabled project settings, maps a causal-loop diagram to identify associated issues and summarises approaches for extracting knowledge from the unstructured content of RFIs. The outcomes of this review stand to offer invaluable insights to both industry practitioners and researchers, enabling and promoting the refinement of RFI processes within the construction domain.

Keywords Request for information, Design coordination, Project performance, Natural language processing, Text mining

Paper type Literature review

1. Introduction

The request for information (RFI) is an important document through which contracting parties seek clarifications in construction documents, including but not limited to plans, drawings, specifications and agreements (Abdel-Monem and Hegazy, 2013). Between the period of design start-up and the call for tender, the design consultants may not have

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adequate time to develop full documents, and as a result, the tender documents are frequently unclear, incomplete, and may cause severe conflicts among various disciplines (Jarkas and Bitar, 2012). Incomplete or ambiguous technical specifications require clarifications, leading to consecutive disruptions to work progress (Jarkas and Bitar, 2012). Also, incomplete detail or lack of clarity in specifications, constructability issues in work packages and differing technical interpretations may result in the burst of RFIs, impeding the smooth flow of work (Jarkas and Bitar, 2012). These practices leave many issues undetected until construction begins, leading to downstream rework. Thus, the RFI process facilitates their timely resolution. Even though RFIs are essential communication tools, they may carry inherent risks (Hughes *et al.*, 2013). RFI is an unavoidable administrative process and a "non-value adding" activity for a construction project (Liao *et al.*, 2020). The occurrence of RFI is unpredictable and cannot be included in a project's baseline schedule. Hence, RFIs have the potential to become part of the critical path, and late RFI response can lead to schedule delays (Kelly and Ilozor, 2020). The processing of reviewing RFI questions and producing a suitable response to satiate the contractor is time-consuming and costly (Love *et al.*, 2014) and may trigger change orders which could become a source of disputes and claims (Hughes *et al.*, 2013). For these reasons, RFIs have been labelled as a "necessary evil" (Aibinu *et al.*, 2019), and therefore, management of RFI needs to be strategised throughout construction projects.

Reducing RFIs and minimising their response time has drawn increasing attention in academic research (Psomas and Alzraiee, 2020; Papajohn and El Asmar, 2021) and industry (Hughes *et al.*, 2013) over the past few decades. While this research direction has generated a growing body of knowledge, the predominant approach embraced by researchers uses human-driven codification (Kim *et al.*, 2021; Morales *et al.*, 2022) to discern patterns within RFIs, which, although insightful, is not pragmatic and applicable in construction projects. In contrast, while the industry has used digital technologies, particularly building information modelling (BIM), to enhance RFI processes, the extent to which these technologies effectively improve the RFI process lacks comprehensive examination. RFIs produce useful, precise and integral project information. Analysing and discussing different themes related to these processes will help the researchers and industry practitioners understand the inherent risks associated with the RFI processes and devise best practices to mitigate those risks. Thus far, there is a lack of a literature review in the body of knowledge that dissects the RFI processes to develop understanding of its intricacies and implications. To foster and direct further research on improving the management of RFI in construction projects, this review paper focuses on several streams. Firstly, this review provides an in-depth insight into the RFI processes and summarises the current key research areas in leading peer-reviewed journal publications from different databases. Secondly, this review will guide researchers in designing future studies and introducing new research avenues to improve RFI management through best practices. The synthesis of the literature on RFI assists the industry in understanding and developing a more robust approach to simplify document-intensive tasks by streamlining RFI processes and minimising human errors. This paper reviews the publications on RFI process and management in the architecture, engineering and construction (AEC) industry for the past two decades (i.e. 2000 to 2023). This period ensures the coverage of advancement and progression of RFI process from traditional to more recent BIM-enabled project lifecycles.

The remaining sections of this review are organised as follows: Section 2 provides an overview of the RFI process in construction projects, while Section 3 explains the methodical approach used to identify and summarise relevant articles. Section 4 discusses the key

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themes in RFI-related research, the contributions of journal publications and current trends in RFI studies. Building upon the identified themes and gaps, Sections 5 and 6 present strategies for efficient RFI management and propose directions for future studies to address research gaps. In Section 7, a comparative analysis and novel insights are provided, situating the present study within the body of knowledge related to construction innovation. Finally, in Section 8, conclusions and contributions are presented.

2. Overview of request for information process

The content of the RFIs is divided into two parts, i.e. structured information and unstructured information body. The structured information includes details such as the project name, RFI number, details of the stakeholders (both initiator and the responder), their company names, the necessary dates, and any supplementary document (CAD drawing or material-related information) supplied as addendum in the following pages. The clarification sought by the contractor and the answer provided by the designer is considered unstructured information (Lee and Yi, 2017). In this process, RFI logs also play a crucial role in facilitating the RFI process. They serve as an administration tool (Papajohn and El Asmar, 2021) to register all the essential details of an RFI in a spreadsheet. The RFI log is a crucial tool to track and monitor the status of an RFI to determine if the status of an RFI is still active or closed.

Typically, the RFI process (Figure 1) starts when an RFI requestor, for instance, a contractor or sub-contractor seeks clarifications to resolve any construction-related problem during the project's lifecycle. Upon receiving the RFI from the contractor, the client/design team registers it into the RFI log before furthering into the review process (Papajohn *et al.*, 2018). The submitted RFIs can be queued and reviewed simultaneously (Chin and Russell, 2008) or may not be

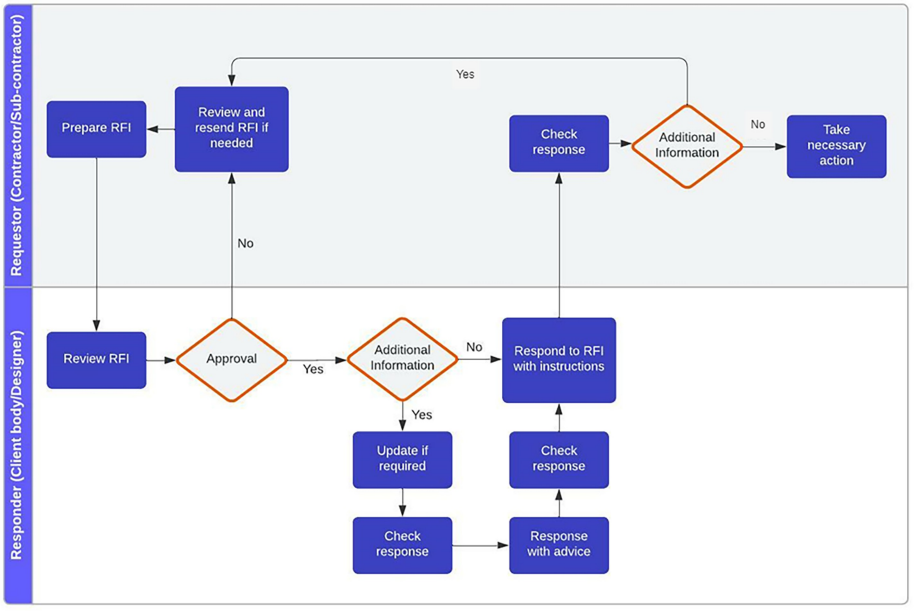


Figure 1.
RFI process in construction adapted from Morales *et al.* (2022)

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queued and addressed independently (Papajohn *et al.*, 2018). Once the responder receives RFI, further actions are taken depending on the clarification sought. Preparing a response to an RFI may trigger further communication sub-process as additional information from other team members and disciplines may be required to address the issue raised (Aibinu *et al.*, 2019; Morales *et al.*, 2022). Thus, responding to an RFI can be hectic and may have cost and time implications. Additionally, excessive delay in RFI response translates to direct demotivation and dissatisfaction among job-site workers, leading to unsatisfactory performance (Jarkas *et al.*, 2014). After analysing the RFI question, a response is formulated, which may trigger further RFIs between the parties involved.

3. Research methodology

This study involves a systematic literature review using a methodology adapted and modified from Zhong *et al.* (2019), comprising four main phases (Figure 2):

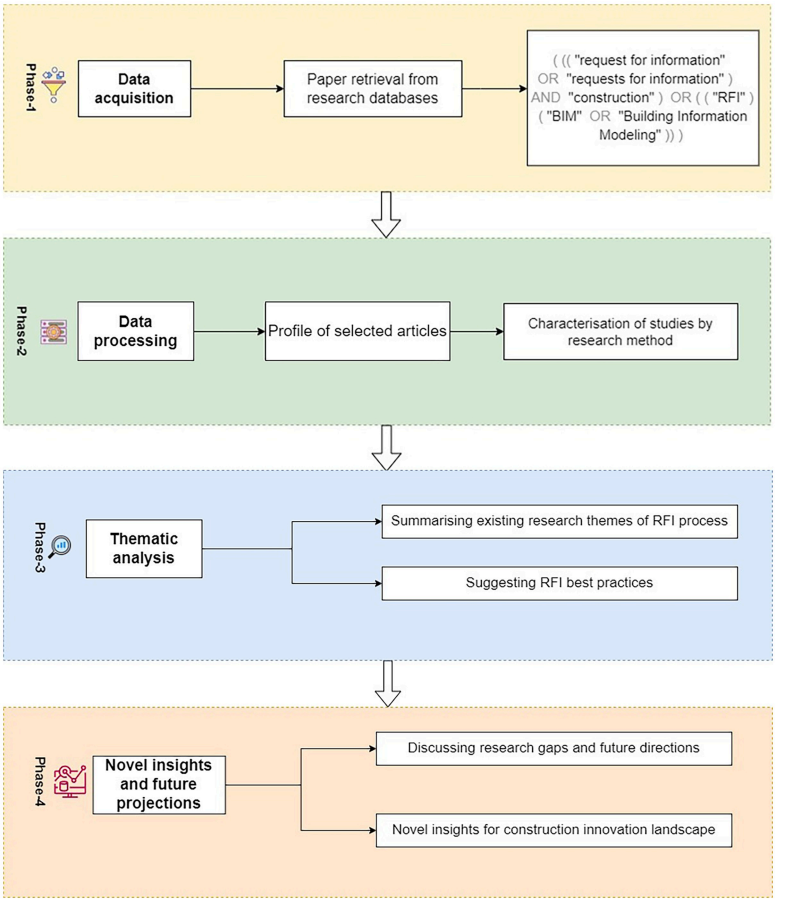


Figure 2.
Outline of research design

Source: Authors' own creation; adapted from Zhong *et al.* (2019)

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(1) data acquisition;
(2) data processing;
(3) thematic analysis; and
(4) novel insights and future projections.

This section discusses data acquisition and processing under their respective subheadings, while separate sections are dedicated to detailed analysis of themes, and future projections. This examination provides valuable insights into best practices, identification of research gaps and suggestions for future developments pertaining to RFI processes within the construction industry.

3.1 Data acquisition: literature search and screening

The literature search and screening framework followed in this study is in line with Baek *et al.* (2021) and consists of three stages:

(1) exhaustive literature search (Stage 1);
(2) duplicate removal (Stage 2); and
(3) qualitative analysis for relevance (Stage 3).

In Stage 1, Web of Science (WoS) and Scopus databases were searched for relevant literature. For this purpose, semantic search strings consisting of RFI-related keywords were developed to retrieve the relevant literature. Different combinations of the keywords joined by Boolean operators of "AND" and "OR" were incorporated. This resulted in 146 articles, with 105 from Scopus and 41 from WoS. From within Scopus results, the articles were restricted to engineering and computer science subject categories to maintain focus and retrieve relevant articles in line with the current research's theme. The default categories for WoS were slightly different; engineering, construction building technology, computer science and business economics were the predominant categories from which the most relevant RFI literature was extracted. The period of these publications was restricted to post-2000, and the language of the articles was limited to English. This period is carefully elected as it encompasses the growing momentum in built environment research from traditional construction management systems towards more machine-dominated or advanced technology-oriented solutions and smart platforms developed by academia and industry (Hooper and Haris, 2010), ensuring maximum research coverage. The studies incorporated in this review included: original research articles, conference proceedings, book chapters and review articles. With an initial investigation and comparison of the research titles in the next stage, 40 studies were removed due to duplication in both databases. In the final step, through initial investigation of the title, abstract and manuscript, 11 articles were removed due to being outside our research scope, leaving behind a final count of 95 relevant articles for the present literature review. Through the implementation of above-mentioned measures, we ensured that our selection comprised exclusively of research articles aligning with our defined scope, specifically focusing on RFIs in the construction sector. These stages of literature screening are illustrated in Figure 3.

3.2 Data preprocessing: characterisation of studies by research method

In this step, a total of 95 shortlisted articles regarding the RFI processes were rigorously reviewed. Each reviewed paper was examined and then characterised by its research method (Li and Kassem, 2021). Within the 95 articles, 44% of the publications gave insight regarding different aspects of the RFI processes and 33% undertook a case study approach

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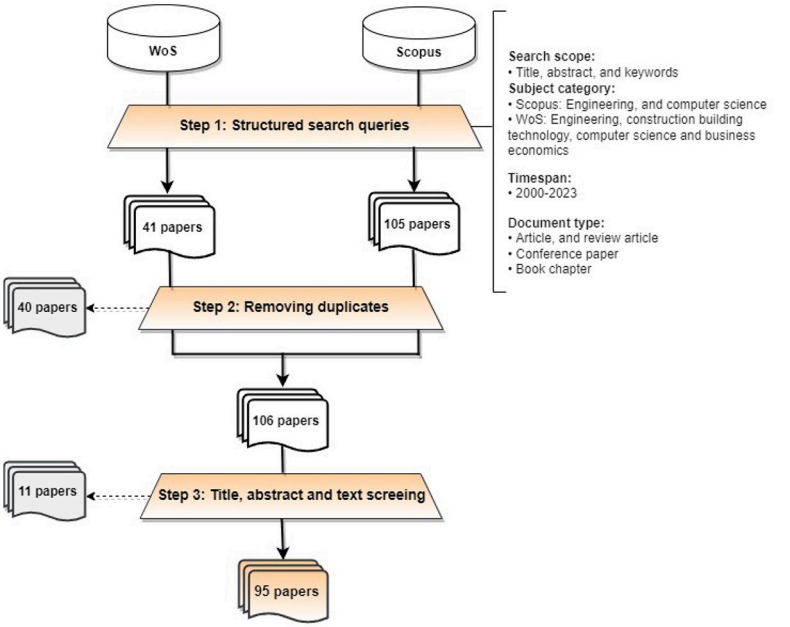


Figure 3.
Literature screening procedure

Source: Authors' own creation; adapted from Baek *et al.* (2021)

and then validated them with some form of experimentation. These experiments can be hypothetical or mimic real-world scenarios and can be demonstrated through existing technological provisions, novel system/prototype or a proposed computational algorithm. Within 31 case study-based publications, the validation of the research was achieved through three approaches, including:

(1) experiment with existing methods/tools (24 articles);
(2) experiment with proposed software system/prototype (four articles); and
(3) experiment with proposed computational algorithm (three articles).

Also, 14% of the total articles aimed at proposing a framework and 7% of the publications incorporated interviews/focus groups/questionnaires/workshops as a research method. Among these articles, 12 research articles proposed a framework, and six publications validated the framework through an experiment. There was only one proof-of-concept simulation article (Papajohn *et al.*, 2018) and one literature review (Sompolgrunk *et al.*, 2021). This review paper briefly discussed RFI as one of the metrics to measure the return on investment for BIM implementation on construction projects. To the best of the authors' knowledge, no comprehensive review has been done to focus on RFI-related research in the construction sector. Therefore, this paper aims to close this gap within the present literature review and investigate the RFI management processes and factors that influence it with a focused scope.

4. Results and discussions

Content analysis is a commonly used research method to analyse and synthesise the research content. In the context of investigating the state of the art in RFI-related literature

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within the construction sector, content analysis was applied to identify patterns, themes and relevant information. Content analysis allows for the collection and organisation of information while revealing trends and patterns present in documents (Krippendorff, 2013). For this study, a combined approach of qualitative and quantitative content analysis was adopted to ensure comprehensive exploration. Qualitative content analysis involves categorising data to gain insights into contextual nuances related to the research. Conversely, quantitative content analysis focuses on deriving numerical values, like frequencies and rankings, by systematically counting occurrences of specific themes or topics (Chan *et al.*, 2009). By using both qualitative and quantitative content analysis, this research seeks to gain a holistic understanding of RFI process in the construction industry.

Through the content analysis of the RFI-related body of knowledge, five main research themes were identified, including:

(1) risk mapping;
(2) influence of project delivery methods;
(3) BIM and RFI management;
(4) Other digital tools to aid RFI management; and
(5) classification frameworks within literature.

Based on these research themes, the study suggests best practices and strategies for managing RFIs, gap analysis, future research directions and novel insights regarding construction innovation knowledge in the following sections.

4.1 Risk mapping

The RFI process is initiated with a contractor formally approaching the designer to seek clarifications. However, this process is characterised as problematic and may lead to project risks. To understand the problematisation associated with the RFI processes, it can be beneficial to apprehend its implications through the systems thinking (ST) approach. ST is a robust method for comprehensively evaluating complex processes (Xu and Coors, 2012). A causal loop diagram (CLD) (Figure 4) is developed to map the interdependencies among challenges mentioned in the literature arising from RFI communication. The development of the CLD involved a comprehensive identification of all causal factors or variables that contribute to the issuance of RFIs. Following the identification of causal factors from the literature, a thorough analysis was undertaken to determine their specific impacts or effects on the RFI process and, subsequently, their influence on the overall project. This step was crucial in understanding how each variable interacts within the system and how changes in one aspect can propagate throughout the project, affecting its outcomes and performance. By examining these relationships, a comprehensive understanding of the broader implications of RFI process on the project's success and efficiency was achieved. The arrows in the CLD indicate the direction of causality, and the linking relationships, whether proportional or inverse, are presented through + or – signs at the tip of arrows, respectively.

Figure 4 shows most arrows concentrating on "RFIs generated" and most arrows emerging from "delay in RFI' response". This is due to researchers heavily relying on these indicators to compare functionality or performance of different project settings. The CLD illustrates that RFIs are generated due to "incomplete contract documents" and/or "mistakes in contract documents" (Gajendran and Brewer, 2012). Contractors and sub-contractors continually discover these discrepancies throughout the project lifecycle (Daoud and Allouche, 2003; Philips-Ryder *et al.*, 2012). In particular, these issues include:

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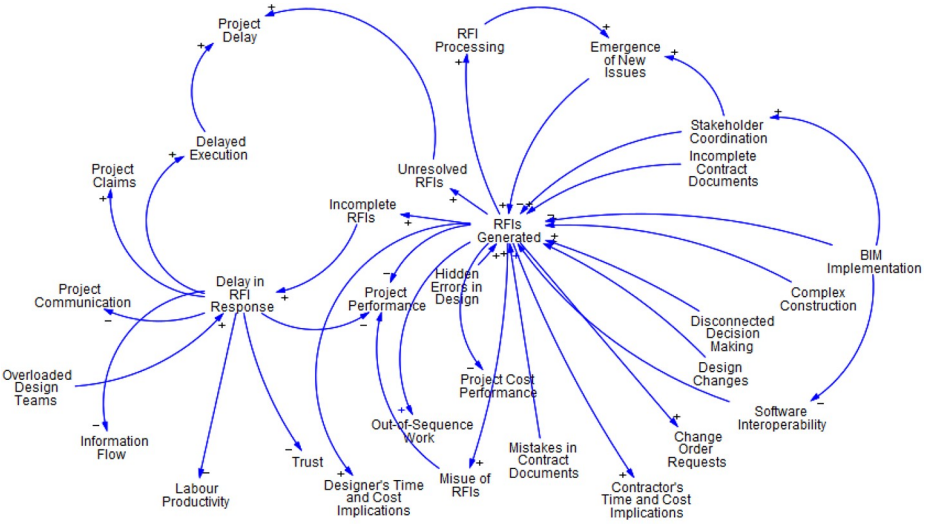


Figure 4.
Causal loop diagram for RFI problematisation mapping

Source: Authors' own creation

- ← imprecise construction documents;
- ← vague specifications;
- ← specification and plan contradictions; and
- ← unanticipated field conditions (Hanna *et al.*, 2012).

If these issues are not dealt with timely, they appear as downstream risks in design omissions, complex designs, uncoordinated drawings, incompatible or inconsistent project specifications (Jarkas, 2015) and field execution problems (Issa *et al.*, 2003). The "emergence of new issues" in CLD gives birth to another cycle of requests for information. The recurrent disruptions also result in low productivity, extended time scales and escalated construction costs. The activities mentioned in the RFIs can lead to project delays and possible claims if they become part of a critical path (Papajohn and El Asmar, 2021). This usually happens due to untimely responses to the RFIs (Papajohn *et al.*, 2018).

Moreover, Figure 4 reveals that "delay in RFI response" is a critical factor that has a detrimental effect on "project communication" (Papajohn *et al.*, 2018) and "labour productivity" (Jarkas *et al.*, 2014). The risks and uncertainties in RFIs, along with resulting schedule delays, can significantly impact the "project cost performance" for both the client and the contractor (Liao *et al.*, 2020). The contractor bears expenses in drafting RFIs and gathering supporting documentation, whereas the client or consultant accrues costs in preparing the response to RFIs (Papajohn and El Asmar, 2021). Furthermore, it is difficult for contractors or sub-contractors to detect informational errors from multiple drawings as it is challenging, time-consuming and error-prone (Filho *et al.*, 2016a, 2016b). Consultants blame inadequate fees and a shortage of skilled workforce to promptly address the RFIs (Philips-Ryder *et al.*, 2012). The CLD also demonstrates that successful "BIM implementation" facilitates enhances "stakeholder coordination" across disciplines during the design phase, resulting in a notable reduction of undetected errors (Filho *et al.*, 2016a, 2016b). Conversely, subpar "BIM implementation" and

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insufficient "software interoperability" (Ali *et al.*, 2014) contribute to a significant increase in RFIs and excessive paperwork burdens.

The developed CLD manifests RFI process as an indicator of "project performance" (Liao *et al.*, 2020), which gets negatively affected by frequent enquiries and design clarifications raised by the contractors. The potential abuse and "misuse of RFIs" is important to note (Hanna *et al.*, 2012). The response to the RFI may also lead to "out-of-sequence" activities (Ibrahim *et al.*, 2020) during execution. Ideally, RFIs should be minimised; however, if used properly, the process may enhance communication and reinforce "trust" between the contracting parties (Philips-Ryder *et al.*, 2012; Zuppa *et al.*, 2016). Overall, this CLD exposes a lack of relationships defined in the literature to develop reinforcing and balancing loops, indicating limited research about RFI problematisation. More sophisticated modelling techniques can be incorporated to create better assessments.

4.2 Influence of project delivery methods

Project delivery systems (PDSs) are defined as the relation between different contracting agencies and the time involved in that relation (Hanna, 2016). The scope of this search includes integrated project delivery (IPD) and traditional PDSs, including design-build (DB), design-bid-build (DBB) and construction management at-risk (CMR). Each delivery method has unique features and varying strengths and discrepancies (Papajohn and El Asmar, 2021). Various researchers have incorporated RFI as an essential metric while comparing multiple project delivery systems. Among other factors, the RFI-related metrics include RFIs issued per million, RFI processing time (Hanna, 2016) and an average cost of an RFI (Kim *et al.*, 2021). Table 1 summarises all the studies in the literature that incorporate different RFI metrics to compare various project settings.

Research study by Bilbo *et al.* (2015) compared CMR and IPD. The CMR project issued 466 RFIs throughout the project, whereas 40 RFIs were issued for the IPD project. El-Asmar *et al.* (2013) report that traditional project delivery methods contain ten RFIs per $1m and a two-week processing time, compared to an IPD project, which has two RFIs per $1m and one-week processing time. Higgins *et al.* (2012) mention that DBB projects lead towards more RFI generation, having less incentive to resolve RFIs compared to methods where contractors are involved right from the design phase. DBB projects fail to achieve an integrated approach because of the incapability to get all the key stakeholders (client, consultants and contractors) to collaborate in the project's planning or design phases (American Institute of Architects, 2007). Contrary to this, (Kelly and Ilozor, 2020) reported no significant differences in RFI metrics for traditional and IPD-based project settings. In the case of PDS with a DB setting, fewer RFIs are expected as they are resolved readily by the DB contractor's engineer [Maryland DOT (MDOT), 2013]. Overall, the RFI metrics are believed to perform better in IPD settings than the traditional PDSs.

Another reason for fewer RFIs in the IPD setting is an agreement between stakeholders to resolve all the issues in a team meeting rather than raising an RFI (Bilbo *et al.*, 2015). Hence, the challenges still exist but are tackled through a different communication channel. In the IPD setting, only issuing an RFI if necessary is encouraged. These measures keep the overall RFI metrics lower. However, there are some disadvantages to the project members in this approach. RFIs serve as an essential tool for record-keeping. In the case of limited RFI, bringing all the project details forward for crucial matters such as approvals and periodic payments becomes challenging. Bilbo *et al.* (2015) refer to a case study wherein IPD setting, more RFIs were produced at the end of the project to generate records and track changes. Furthermore, IPD only effectively reduced the number of formal RFIs; however, the project delivery method did not affect the quantum of issues/disruptions raised on the project.

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Table 1.
RFI Metrics in traditional PDSs and IPDs

Delivery method (s)	Project type	Research scope (RFI-related)	Total no. of projects	BIM-enabled	RFI-related metric	Unit	Findings	Citation
CM/GC	Highway	Determining impact of CM/GC project delivery on RFI response time	1	N/A	RFI response time	# of construction days delayed	During the project, 66 construction days were taken to respond to RFIs The days delayed may or may not translate to project delay	(Papajohn *et al.*, 2018)
DBB; CM/GC; DB	Highway	Comparing the impact of DBB, CM/GC, and DB on RFI response time	17	N/A	RFI response time	# of construction days delayed	No. of days delayed to produce response during the project are below: • DBB: 139 days • CM/GC: 125 days • DB: 380 days The days delayed may or may not translate to project delay	(Papajohn and El Asmar, 2021)
IPD	Energy	Developing IPD implementation framework for energy project	1	✓	RFI frequency	# of RFIs	No mention of no. of RFIs The framework reduced the project cost by 20% and cut project duration by 25%	(Psomas and Alzraiee, 2020)
IPD; CMR	Health care	Comparing the impact of IPD and CMR on RFI frequency	2	IPD (✓); CMR (N/A)	RFI frequency	# of RFIs	For both projects the number of RFIs were lower than expected average due to great working relations between the teams • CMR: 466 RFIs9 • IPD: 40 RFIs	(Bilbo *et al.*, 2015)
IPD; traditional PDSs (DBB, CM	Varying	Comparing impact of IPD and traditional	93 (18 IPD)	N/A	RFI frequency	# of RFIs	Overall, at the 95% confidence level, no significant RFI differences between IPD and	(Kelly and Ilozor, 2020)

(*continued*)

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Delivery method (s)	Project type	Research scope (RFI-related)	Total no. of projects	BIM-enabled	RFI-related metric	Unit	Findings	Citation
GMP; CM cost +← and DB)		PDSs on RFI frequency					traditional PDSs IPD health-care projects had significantly fewer RFIs than non-IPD healthcare projects	
IPD; Non-IPD (CMR, DBB and DB)	Infrastructure	Comparison of IPD and non-IPD on RFI frequency and response time	N/A	N/A	RFI frequency; RFI response time	# of RFIs/ million dollars; construction weeks	RFI response time • IPD: 1.4 weeks; Non-IPD: 1.5 weeks RFI frequency • IPD: 3.9 RFIs/million dollar; Non-IPD: 8 RFIs/million dollar	(Hanna, 2016)
DBB; DB	Infrastructure	Comparing cost performance of DBB and DB with RFIs	2	N/A	Avg. cost/ RFI	$	Average cost/RFI: Design-build method • Owner: $1,360; Builder: $4,018 Design-bid-build method • Owner: $23,555.60; Builder: $783.02	(Kim *et al.*, 2021)

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Table 1.

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Hence, project clients should also reassess their unrealistic expectations of IPD as a solution to construction problems, as the trade and qualitative literature suggested.

Irrespective of the project setting, RFI communication is an administrative process controlled and processed within an office setting. Hence office-based RFI communication is critical to on-site execution, and improving office-related processes can streamline RFI metrics (Alves *et al.*, 2016) and enhance process management. Communication sequences for RFIs and submittals must be adequately managed and evaluated based on project characteristics, constraints and needs to avoid delays (Alves *et al.*, 2016). To intelligently manage RFIs, it is necessary to understand actions that might impede communication sequences and avoid automating wasteful practices. IT provisions for RFI processes should be linked with construction documents and schedule activities to provide managers with a more comprehensive view of their RFIs and necessary information to close active RFIs. The system can highlight which RFIs need urgent action or remain overdue and send alerts to those accountable for closing the RFIs.

Finally, managing RFIs during construction projects is an inherent challenge irrespective of the project delivery method adopted and the literature reports using digital design technologies facilitating the RFI process. The following section discusses the merits and demerits of BIM for RFI management.

4.3 Building information modelling and request for information management

4.3.1 Strengths of building information modelling to support request for information process. Reduced number of RFIs in BIM-enabled projects is one of the top-mentioned benefits in the body of knowledge (Barlish and Sullivan, 2012). BIM implementation can improve the overall design by significantly improving interdisciplinary design coordination, which aids in reducing the number of requests for information (Filho *et al.*, 2016a, 2016b). Contrary to a BIM-enabled lifecycle, in a non-BIM setup, construction professionals cannot readily visualise conflicts through 2D drawings, resulting in an administrative burden to resolve (Filho *et al.*, 2016a, 2016b). Unmanaged conflicts at the design stage led to rework and a burst of RFIs, which inhibit project productivity. BIM pinpoints the likely problems before the beginning of construction work providing a robust mechanism to deal with RFIs. Filho *et al.* (2016a, 2016b) explain the BIM-based RFI management process. Firstly, BIM-based models relevant to different disciplines are generated. Secondly, constructability analysis is performed through a 3D-coordinated/federated model containing all designs; a report with all RFIs is generated. The design teams perform the reviews and release the updated version of the models. The coordinated model and latest design are examined in the final step to ensure that pointed problems have been solved. In this stage, knowledge exchange takes place among various disciplines and virtual, design and construction teams to ensure that the design is validated, and most RFIs have been resolved. After modelling the design changes, a final report is released containing comments unresolved. Giel and Issa (2013) explain in detail how BIM-enabled design can reduce the number of RFIs:

- ← Reduced dimensional inconsistencies: BIM helps reduce discrepancies and rounding errors in 2D drawing sheets.
- ← Reduced document discrepancies between disciplines: The information segregation during 2D design drawing preparation in each discipline leads to errors that are more significant. BIM ensures smooth design coordination across different disciplines reducing the number of RFIs.

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- Reduced 2D errors and omissions: BIM provides a virtual model at the pre-construction stage, which provides a chance to minimise the clarifications that otherwise become RFIs.
- Reduced grid or column alignment issues: During design, column and grid alignment contributes to major source of conflicts in a project; however, in BIM-enabled design, these issues are resolved during pre-construction, reducing the occurrence of RFIs.
- Reduced direct clashes: Advanced visualisation abilities in BIM resolve conflicts between systems and disciplines early. These conflicts may include clashes between architectural, structural, HVAC and other building systems.

4.3.2 Challenges in building information modelling-based request for information management. Although BIM has drastically improved the landscape of AEC industry in terms of design and execution, research has pointed to discrepancies and inefficiencies related to issue handling and RFI management. At the award of contract stage, the general contractor seeks clarifications from consultants to finalise their BIM models (Liao *et al.*, 2020). Eventually, the contractor must re-create the MEP design models due to insufficient documents from designers. During the design phase, due to poor BIM collaboration among different services, design issues are not resolved until the execution, resulting in a large number of RFIs, and a potential project delay (Liao *et al.*, 2020). Poor BIM management practice creates excessive paperwork for all the parties involved in the project.

Similarly, architects and engineers take long to respond to contractors' RFIs as their design models do not directly guide site work (Liao *et al.*, 2020), causing delay risk. With such a scenario at hand, the speciality contractors are also left with inconsistent and uncoordinated models and drawings and unresolved clashes to carry out execution activities to manage the tight construction schedule models (Liao *et al.*, 2020), starting the vicious cycle of RFIs. Furthermore, trade contractors rarely incorporate BIM (Lam, 2014); this practice leads to executing construction activities that are unplanned and inconsistent with the design models of the in-house BIM. Moreover, clashes detection during execution leads to abortive works, resultantly requiring extra time and person-hours to re-design, re-do and rebuild the construction work (Liao *et al.*, 2020). These issues can be streamlined by developing a BIM execution plan (BEP) that properly narrates how federated models will be updated to showcase the final design intent. Further, BEP should include the frequency and extent to which change orders and RFI responses affect design intent and should be incorporated into the design authoring models.

Despite the overall decrease in RFIs due to improved visualisation and experiential experience, the nature of the RFIs generated as a result of poor BIM coordination practices and BIM-based design deficiencies remains unclear, primarily due to the absence of research on this aspect (Liao *et al.*, 2020). There is also a relationship between BIM professionals' skill level in design modelling and on-site project productivity based on rework and RFIs (Fan *et al.*, 2014). If the BIM modeller does not incorporate the required level of detail for the speciality trade, then BIM will not serve as an accurate virtual representation of the on-site components. Hence, inexperienced BIM modellers become the primary reason for BIM failure, which increases the amount of rework in the field and the number of related RFIs (Fan *et al.*, 2014). Moreover, with BIM in practice, not all the stakeholders have the same level of information, especially trade contractors. This gap in knowledge between stakeholders becomes the source of errors during design and construction, leading to the vicious cycle of RFIs. To assuage these negative implications, BEP should narrate a framework acting on which team members determine whether an RFI response disturbs

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design intent and a mechanism to track any changes in design authoring models that arose due to these RFI responses. Considering the limitations of BIM for RFI management, both industry and academia have explored other technologies to streamline the process, which are discussed in the following section.

4.4 Other digital tools and platforms to aid request for information management

The construction industry has observed a subtle shift in managing RFIs from paper/email-based channels to online. Table 2 points out the technologies mentioned in the literature to aid different aspects of the RFI processes. Currently, several electronic data management systems (EDMS) are available that allow construction stakeholders to manage their RFI documents (Kähkönen and Rannisto, 2015). EDMS are Web-based platforms that enable tracking changes and traceability (Erri Pradeep *et al.*, 2021) for RFIs. This makes their usage and applicability more efficient than traditional paper/email-based RFI communication, lacking data traceability. With the advent of BIM in the construction industry, more sophisticated data management provisions are now available. A common data environment (CDE) is one such platform that facilitates model-based project management (Erri Pradeep *et al.*, 2021). CDE platforms that support RFI administration and handling include Aconex (Das *et al.*, 2020; Aibinu *et al.*, 2019), Autodesk 360 (Das *et al.*, 2020) and Procore. CDE provides a collaborative platform for project stakeholders to exchange information through cloud technology. Both EDMS and CDE have helped the construction industry handle RFIs; however, they present a few challenges. Some of these challenges include; complex data

Technology category	Technology/software	Functionality to aid RFI management	Citation
Common data environment	Aconex, Autodesk 360	RFI administration/ management/handling	Das *et al.*, 2020 (✓)
	Aconex		Aibinu *et al.*, 2019
Emerging technologies	Blockchain	RFI administration/ management/handling	Das *et al.*, 2020
		Information exchange and data security	Erri Pradeep *et al.*, 2021 (✓)
ERP system	RFI automation tool	RFI administration/ management/handling	Ekstrom and Bjornsson, 2005
Immersive technologies	Mixed reality	Design coordination and collaboration	Alizadehsalehi *et al.*, 2019 (✓)
Other ICT provisions	Interactive voice response, email	Communication	Abdel-Monem and Hegazy, 2013
Software solution	Unity 3D, 3DS Max	Design visualisation	Castronovo *et al.*, 2018 (✓)
	Unity 3D	Design coordination and collaboration	Alizadehsalehi *et al.*, 2019 (✓)
	CMIC	RFI administration/ management/handling	Sawyer, 2007
Text mining	Metadata modelling	Content analysis	Zhu *et al.*, 2007; Mao *et al.*, 2007
	Topic modelling	Risk assessment	Lee and Yi, 2017
	QCA software	Content analysis	Soh *et al.*, 2020
Web-browser interface/ Web-based platform	Electronic data management system (EDMS)	RFI administration/ management/handling	Kähkönen and Rannisto, 2015 (✓); Papajohn *et al.*, 2018

Table 2.
Technologies related to RFI management

Note: (✓) = BIM-enabled
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architecture, storage problems, interoperability issues within EDMS (Kähkönen and Rannisto, 2015) and data loss and legal issues within CDEs (Erri Pradeep *et al.*, 2021). Academic researchers are now embracing emerging technologies to develop transparent and secure file handling platforms to deal with these challenges. Studies by Das *et al.* (2020) and Erri Pradeep *et al.* (2021) provide a decent start towards developing Blockchain-backed prototypes to develop better RFI management systems. However, these studies still require industry validation to substantiate the results claimed in the studies.

Since RFIs comprise unstructured text, different researchers have explored the text mining approach to decode the content of the RFIs. In this regard, Zhu *et al.* (2007) and Mao *et al.* (2007) developed metadata models to facilitate the processing of the RFIs. Lee and Yi (2017) recently used text mining through the topic modelling approach and categorised RFIs based on their risks. Researchers have also used immersive technologies to improve various aspects of design, such as design visualisation, coordination and collaboration to proactively detect the issues, which may later become part of the RFI process. Alizadehsalehi *et al.* (2019) focused on mixed reality. They integrated it with BIM to enhance stakeholders' coordination which indirectly leads to timely resolution of issues, ultimately reducing the negative implications of the RFI process. Most of the approaches mentioned in the literature had an indirect impact on the RFIs and limited to no research aims to reduce the review time of RFIs using technologies.

4.5 Classification methods within the literature

RFI has been criticised for the lack of structure (Mao *et al.*, 2007). Unstructured information presented in RFI can be related to any issue, depending on the clarification by the contractor. Different researchers have characterised the reasons or causes of RFIs to analyse their content. Through this analysis, industry stakeholders can delve into important questions such as:

- ← why the issue originated; and
- ← how it could have been addressed (Bhat *et al.*, 2017).

Based on these explorations, best practices and strategies can also be devised to reduce the turnaround time and improve the overall RFI process (Kim *et al.*, 2021). Considering the anatomy of an RFI, contractors or owners do not classify the RFIs based on the issue/reason codes (Kim *et al.*, 2021); hence researchers have developed different classification methodologies to categorise RFIs and analyse their content. Figure 5 provides an alluvial diagram representing different classification frameworks and the sub-classifications incorporated in the literature. Past research has classified RFIs based on the property code, discipline code, work-element code, category/type of RFI and reason/issue behind RFI. The individual node height is an indicator of frequency. Within these categories, classification based on the category/type of RFI and reason/issue has been the predominant categories. Each researcher has categorised the RFIs based on their own experience and knowledge (Kim *et al.*, 2021). Hence, it can be observed that few RFI sub-classifications belong to two different categories. For example, design clarification is termed as a type category by Bhat *et al.* (2017) and, at the same time, an issue category by Hanna *et al.* (2012).

Human driven codification highlights multiple sub-classification categories, such as alternative design solutions, approvals, confirmation, conflict, construction coordination, design coordination, design modification, value engineering, constructability issues, differing site conditions, omission, divergence and design clarification, as predominant and recurring issues within the RFIs, indicating their significance in construction projects. Interestingly, few researchers have incorporated multiple classification methods to analyse

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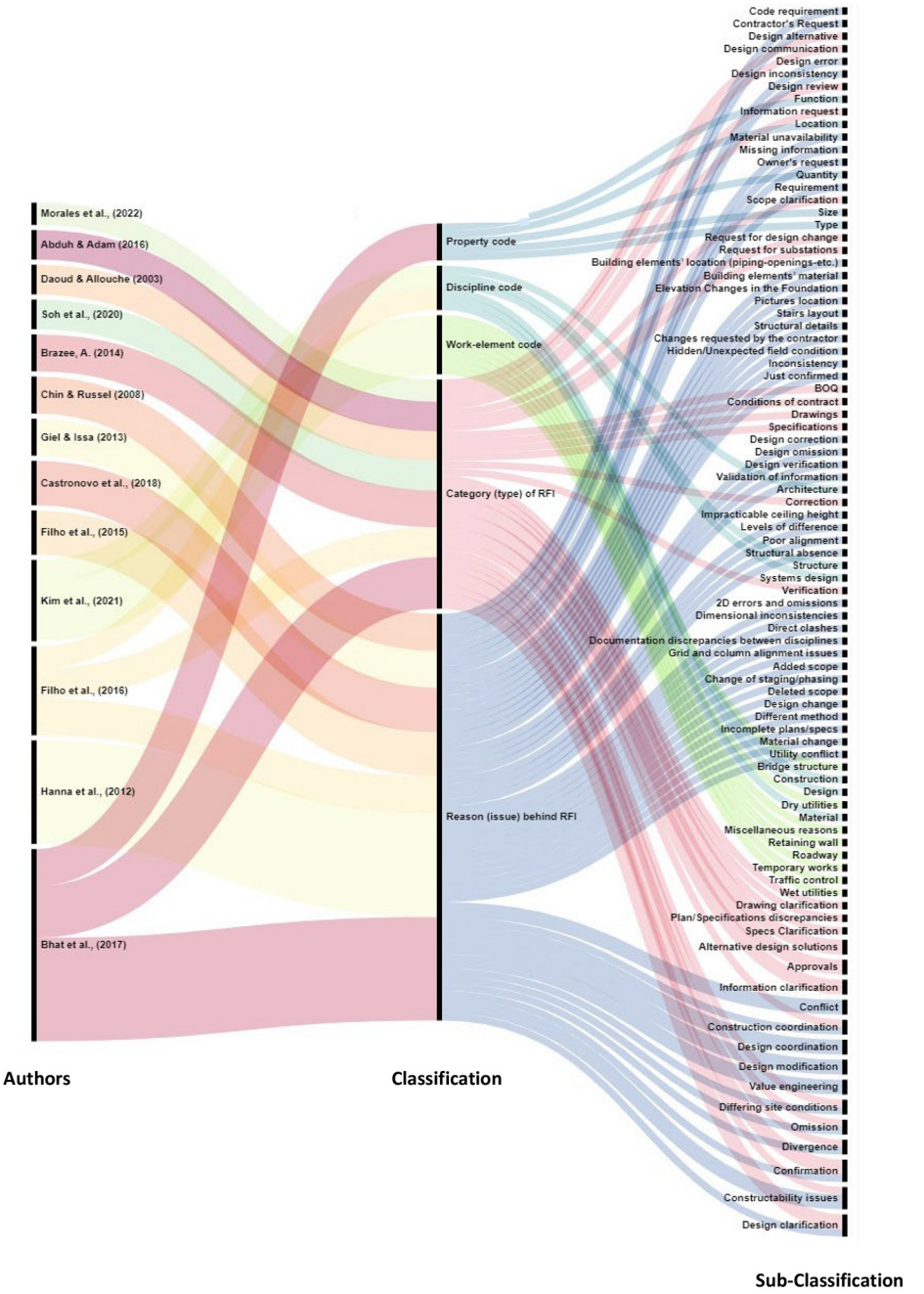


Figure 5.
RFI classifications mentioned in the literature

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the RFI documents. For example, Filho *et al.* (2016a, 2016b) devised a two-stage RFI classification methodology. Firstly, the RFI was classified based on correction, omission, verification and divergence categories. Then, the authors reposed a coding process to organise the RFIs into issues, such as, conflicts, poor alignment, impracticable ceiling height, level differences and absence of structural elements as the predominant problems. Similarly, Bhat *et al.* (2017) followed a three-step codification process to annotate and analyse the RFIs. These classifications can be valuable to the project team as they enhance communication, navigate the project team and align the project towards intended and desired results (Brazee, 2014).

5. A way forward on request for information management

With the viewpoints taken from the above themes, the following five groups discuss strategies and best practices for the better overall management of the RFI process that can potentially reduce the number of subsequent RFIs:

(1) *Effective communication:* Clear expectations of the project outcomes should be defined at earlier stages by the owner (Liao *et al.*, 2020). Otherwise, unexpected scope deviations at later stages delivered through RFIs and other project documents make difficult for contractors to execute within established time and budget. Hence, project programme meetings should be conducted, integrating them with the RFI management system to share information, foster meaningful exchanges between teams (Hanna *et al.*, 2012) and promptly discuss strategies for swiftly closing active RFIs.

(2) *Standardised RFI process:* RFIs should be prepared in a standardised format with a proper mentioning of elements, such as; number and date, the identity of the author, a reference to the design or specification document, suggested solution, response time, a tracking number and an indication to schedule and cost impacts (Hanna *et al.*, 2012). Hanna *et al.* (2012) further add; that an RFI should seek only one clarification, ideally with a probable solution to decrease the response time. Batching RFIs is considered malpractice (Andrews, 2005). From consultant's/owner's end, the response should be timely and guide towards resolving the problem. In case of a delay, the submitter should be notified. Furthermore, the RFI logs should be updated consistently, and in case of any change, they should be communicated to all the concerned parties (Andrews, 2005). A change order request should be initiated if an RFI changes the scope of work or a change in contract conditions.

(3) *BIM-based measures:* For the projects incorporating BIM, to manage any RFIs that are generated due to the non-standardised BIM-design coordination, a contractual framework should be established that promotes RFI-based collaboration (Piroozfar *et al.*, 2019), especially for speciality designers (Morales *et al.*, 2022). This minimises errors (Liao *et al.*, 2020), raising fewer RFIs. BIM implementation plan should incorporate a digitised RFI management system that enforces all the stakeholders' participation from the earlier design stages (Alreshidi *et al.*, 2017). The lack of skilled BIM labour (in speciality contractors) is an issue that hampers desirable BIM implementation. RFIs that are caused due to such modelling mistakes can be managed by frequent BIM-QA/QC and code compliance checks (Donato *et al.*, 2018). Further, the client body can arrange special training to upskill the team. This way, additional costs, duplicate efforts, and RFIs generated due to poor BIM modelling practices can be minimised.

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(4) *Realistic timelines for design:* Insufficient durations should not be imposed upon designers to develop and specifications, and tender documents and review design alternatives. This leads to design documents remaining often ambiguous, incomplete and containing myriad of unresolved issues within disciplines (Daoud and Allouche, 2003). An accurate schedule ensures the timely completion of design works. On the contrary, an unrealistic schedule may backfire on the design team and the quality of design packages. The time saved in the design phase could accrue several times during bidding and execution (Daoud and Allouche, 2003). Incorporating lessons learnt from findings of previous projects can provide an understanding of the mistakes that can be avoided when designing for future projects.

(5) *Constructability exercises:* The design firms should actively engage in constructability exercises to avoid consecutive interruptions to the work progress. Sorting errors initially reduces the number of RFIs; otherwise, the same omissions exacerbate the morale of workforce, creating frustration, demotivation and low productivity. Jarkas *et al.* (2015) and Papajohn and El Asmar (2021) suggest that the appointment of a high calibre project manager at earlier stages can ensure; scrutiny of contract documents (Hanna *et al.*, 2012; Daoud and Allouche, 2003), detection of pitfalls, and missing details and eventually prepare an in-depth RFIs to avoid downstream risks.

6. Research gaps and future studies

This review identifies gaps in RFI management research and proposes further investigation to provide solutions and enhance understanding.

6.1 Improved information exchange platforms

In this review study, several shortcomings of existing RFI management platforms were identified, including:

- ← complex folder structure;
- ← slow and ineffective system usage;
- ← combination of documents and digital data; and
- ← different end-users and project needs.

This problem becomes more complex in BIM-enabled projects. BIM standardisation and software interoperability remain the challenges encountered by industry and academic. Many speciality contractors and suppliers do not have appropriate BIM suites and skilled labour to operate efficiently. Furthermore, the specific RFIs arising from BIM-related issues have not been extensively explored. The optimal data exchange will happen only if the design team, the prime contractor, all subcontractors and suppliers/vendors have full access to the BIM documents and use the same BIM software packages along with efficient platforms that streamline RFI sharing, tracking and monitoring.

6.2 Automated approaches to determine request for information priority

Streamlining RFIs requires prompt responses from consultants/owners to avoid negative impacts on the project schedule (Papajohn and El Asmar, 2021). RFIs should indicate the date and priority of the requested clarification, as prioritisation can aid in timely responses

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and avoiding delays (American Institute of Steel Construction, 2016). A priority provided by the contractor will be backed by approximate estimations. Predictive data analytics can assist in determining RFI priority, allowing for allocation of resources and timely closure of urgent RFIs, thus avoiding disruption to the project's critical path.

6.3 Analysing request for information content through text mining

Analysing RFI content is essential for learning from past experiences and enhancing design practices for future projects. To achieve this, advanced techniques such as text mining and natural language processing (NLP) must be used to comprehend the unstructured content within RFIs. Natural language processing is a realm within artificial intelligence that aids in dissecting, comprehending, and manipulating human languages, encompassing unstructured text documents as well as human speech (Hassan and Le, 2021). Leveraging these approaches will provide valuable insights from RFIs, enabling the identification of various risks. Consequently, this will enhance the handling of conventional RFI documents and reduce the turnaround time of RFIs. However, the reason this type of research has not been extensively explored yet is because the adoption of advanced text mining approaches in the construction field is progressing slowly. Extracting knowledge from construction documents using text mining remains a challenging task. Additionally, the sensitivity of RFI documents makes it even more difficult to access relevant documents, as they often reveal project mistakes and are not readily available.

6.4 Critical risk factors identification

This review paper developed a preliminary CLD to map different challenges associated with the RFI process. However, it serves as a qualitative conceptualisation due to the lack of relationships mentioned in the literature. Future research may incorporate industry feedback to develop a systems dynamic (SD) model to establish the missing links in the literature between different variables through quantitative analysis. A robust SD model will serve both body of knowledge and industry professionals, a holistic understanding of the causality relationships within the RFI process.

7. Comparative analysis and novel insights: situating the review within the construction innovation landscape

The present study aligns with literature reviews on NLP in construction, revealing limited research on leveraging NLP and text mining to extract valuable insights from construction documents (Wu *et al.*, 2022). Additionally, the scarcity of research on applying text mining techniques to construction documents underscores data sharing challenges due to sensitive project information (Yan *et al.*, 2020; Baek *et al.*, 2021). Notably, this literature review advances construction innovation knowledge by raising awareness of ongoing trends and emphasising the need for text mining and NLP techniques to extract valuable information from unstructured construction documents. Figure 6 offers valuable insights into the diverse applications of NLP and text mining for construction documents, showcasing the multitude of benefits that can be obtained by embracing these advanced approaches.

Distinctively, previous research lacks a comprehensive examination of the evolution from traditional construction processes to advanced technologies, such as BIM, while advocating for analytics integration. For instance, a specific article by Sompolgrunk *et al.* (2021) focuses on BIM's impact on the RFI process, while research by Rowlinson (2017) and Karasu *et al.* (2022) delve into BIM implementation within IPD settings. Similarly, Ali *et al.* (2022) attempt to address construction claims but limit their suggestions to BIM and other digital technologies, overlooking the transformative potential of advanced analytics in text

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mining research within the construction domain. Unlike the saturated body of knowledge surrounding BIM-related research (Abdal Noor and Yi, 2018), this study provides a fresh perspective.

A pioneering aspect of this research is its comprehensive examination of the entire RFI lifecycle, from traditional practices to BIM-enabled processes and beyond. Such a holistic approach makes it a trailblazer in the construction innovation landscape. The present literature review focuses on a critical construction communication process, thoroughly dissecting its limitations, technological advancements and future trajectory for further enhancement. Emphasising the adoption of recommended strategies, the potential benefits are substantial, ranging from driving advanced practices to optimising project resources and enhancing construction business profitability. This systematic research approach and extensive coverage of RFI processes has the potential to lead and serve as valuable roadmap for other construction specific documents or processes. By adapting the methodology and insights derived from this research, other construction-related documents, such as change orders, contract documents, specifications, claims and request for proposals, can be subject to similar analyses. The integration of text mining and NLP methods, as briefed in this study, can reveal actionable insights from construction documents, leading to improved decision-making, streamlined processes and desired project outcomes.

In addition, this research recognises the transformative potential of intelligent large language models such as ChatGPT, that can streamline knowledge extraction from construction documents. Embracing these technologies can revolutionise the construction industry, offering efficient data analysis and improved decision-making. In conclusion, the adoption of cutting-edge technologies offers transformative possibilities for the industry, leading to increased efficiency, better project outcomes and heightened competitiveness. As the construction landscape evolves, embracing advanced text mining and NLP will play a pivotal role in shaping the future of construction innovation.

8. Conclusion

This literature review serves as a comprehensive review of existing studies on RFI process in construction industry. A number of themes in RFI management literature were identified and discussed in this study, including:

- risk mapping;
- influence of different project delivery methods;
- BIM and RFI management;

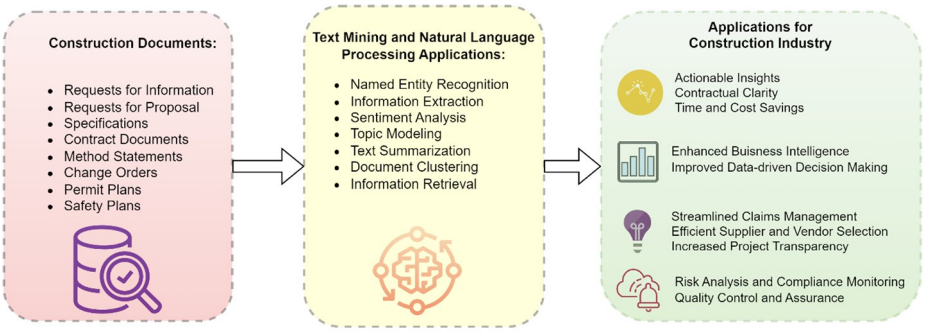


Figure 6.
Harnessing NLP and text mining for enhanced construction document analysis

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- other digital tools and platforms to aid RFI management; and
- classification methods within the literature.

It is revealed that the RFI processes directly impact project interface, success and productivity. RFI frequency and response time have been regarded as essential metrics to compare different project settings. In this review study, it also found that no project delivery method can be deemed most suitable for minimising the risks due to the RFIs.

Overall, this literature review serves as a comprehensive reference for academicians and industry experts interested in the RFI processes in the construction sector. For academia, this study provides a holistic understanding of the RFI process by discussing its repercussions in both traditional and BIM-enabled lifecycles. The CLD diagram was developed to illustrate the causal relationships between different project parameters that influence the RFI process. Further, this study summarises all the classification frameworks developed by different researchers to decode the unstructured content of the RFIs. Based on the discovered themes and critical literature analysis, this study presents the best practices and strategies. This review extends the body of knowledge and may help world of practice in:

- reducing the occurrence of RFIs; and
- in case of such events, recommend measures to reduce the RFI review period.

While existing research has attempted to categorise and classify the RFIs to understand the root cause behind an RFI, it is time-consuming to manually develop and implement these classification methods. Further studies are recommended to develop a smart and efficient approach to categorise and classify RFI and minimise the errors caused during the RFI analysis. Future research could utilise text mining and NLP methods to rapidly generate quick, actionable insights for industry stakeholders to review and close an RFI. Moreover, existing CDE platforms should also incorporate analytics so that the accumulating data guides towards efficient management of RFIs for future projects.

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